



EG1113 - C Programming for Engineering Lab

Report #7

Tesla Mode

Date of the experiment: October 9, 2024

Due date of report: October 11, 2021

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1. Core concepts learned

- Setup our camera to stream to the web
- Track an object via the camera and utilize as an analog input

2. List of equipment and software used

- Raspberry PI 400
- Notepad
- Lenovo Laptop
- Caja File Network Share

3. Relevant web links and other documents:

- Utilized [Raylib](#) to create visual graphics with the PI and to collect input from key presses

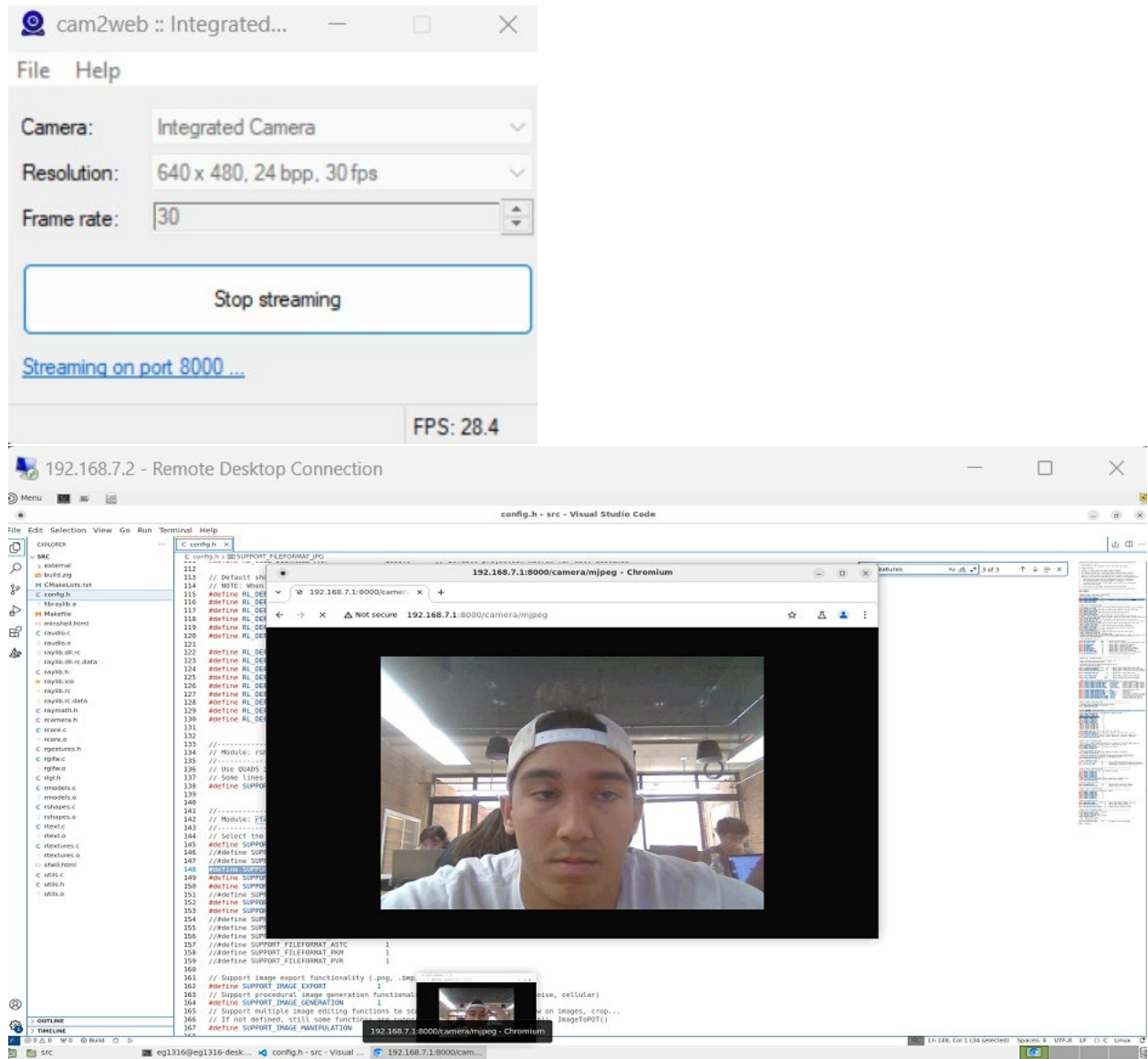
4. Exercise Modify Raylib & Setup Web Camera

4.1 Summary

- Setup Cam2Web application to track an object with the pi

4.2 Procedures

1. Install and run Cam2Web, verifying it works on PI (lab verification)



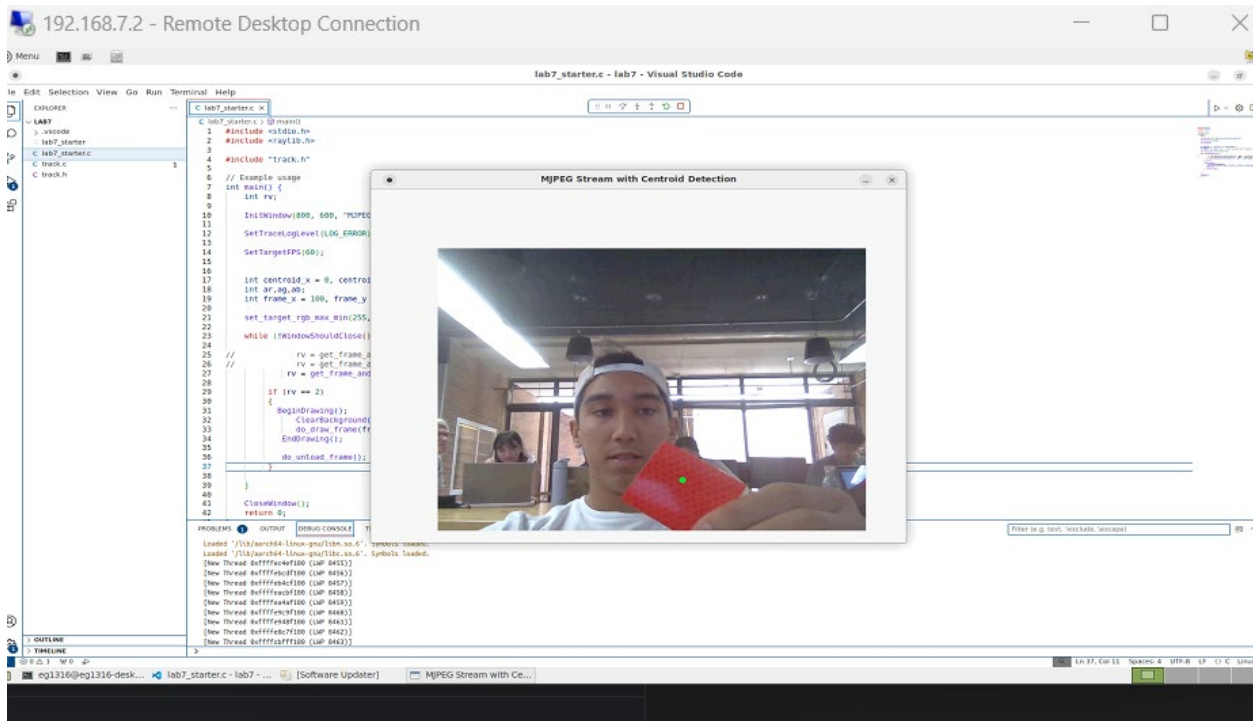
2. Modify and apply changes to raylib

```

340
341 //-----
342 // Module: [Textures] - Configuration Flags
343 //-----
344 // Select the desired fileformats to be supported for image data loading
345 #define SUPPORT_FILEFORMAT_PNG 1
346 #define SUPPORT_FILEFORMAT_BMP 1
347 #define SUPPORT_FILEFORMAT_TGA 1
348 #define SUPPORT_FILEFORMAT_JPG 1
349 #define SUPPORT_FILEFORMAT_GIF 1
350 #define SUPPORT_FILEFORMAT_DDS 1
351 // #define SUPPORT_FILEFORMAT_PSD 1
352 #define SUPPORT_FILEFORMAT_DOS 1
353 #define SUPPORT_FILEFORMAT_HDR 1

```

3. Verify that track (lab7 starter code) works



5. Exercise Tesla Mode

5.1 Summary

- Modify the code to support a tesla mode that follows an object
- Lab verification: [lab 7](#)

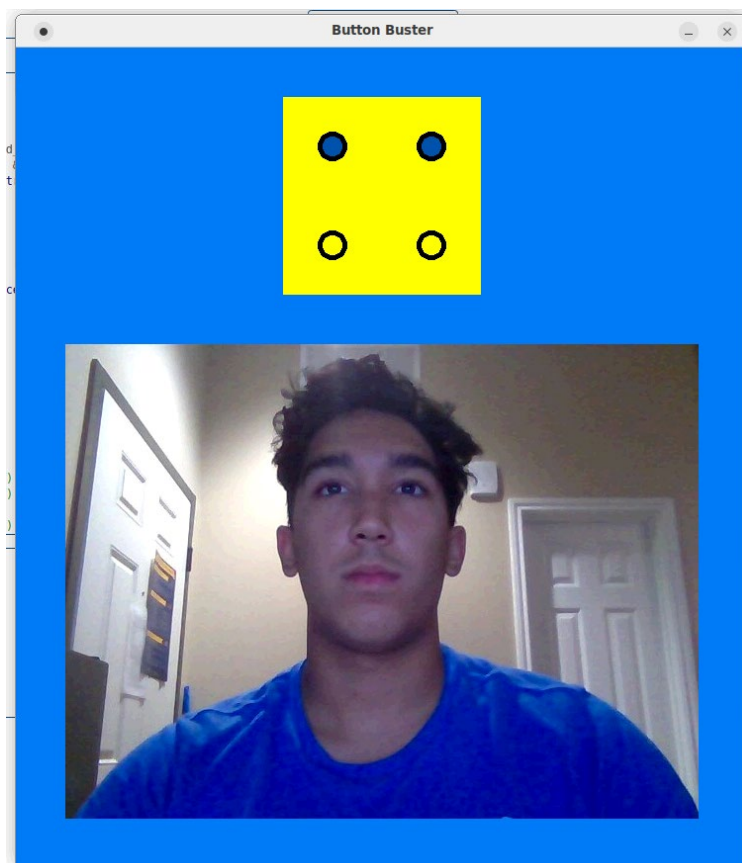
5.2 Procedures

1. Modify the code to display both the track and buttons on computer display

```

30 void draw(int buttons)
31 {
32     // rv = get_frame_and_centroid("192.168.1.122", "8080", "/?action=stream", &centroid_x, &centroid_y, &centroid_weight, &ar, &ag, &ab);
33     // rv = get_frame_and_centroid("10.211.55.25", "8080", "/camera/mjpeg", &centroid_x, &centroid_y, &centroid_weight, &ar, &ag, &ab);
34     rv = get_frame_and_centroid("192.168.7.1", "8080", "/camera/mjpeg", &centroid_x, &centroid_y, &centroid_weight, &ar, &ag, &ab);
35
36
37
38     if (rv == 2)
39     {
40         //ClearBackground(RAYWHITE);
41         do_draw_frame(frame_x, frame_y, centroid_x, centroid_y, (IsKeyDown(KEY_TESLA_MODE) ? centroid_weight : 0));
42     }
43
44     DrawRectangle(center_x - SQUARE_SIZE / 2, 50, SQUARE_SIZE, SQUARE_SIZE, YELLOW);
45
46     DrawCircle(center_x - SQUARE_SIZE / 2 + 50, 100, 15, BLACK);
47     DrawCircle(center_x + SQUARE_SIZE / 2 - 50, 100, 15, BLACK);
48     DrawCircle(center_x - SQUARE_SIZE / 2 + 50, SQUARE_SIZE, 15, BLACK);
49     DrawCircle(center_x + SQUARE_SIZE / 2 - 50, SQUARE_SIZE, 15, BLACK);
50
51     DrawCircle(center_x - SQUARE_SIZE / 2 + 50, 100, 10, buttons & 8 ? BLUE : DARKBLUE);
52     DrawCircle(center_x + SQUARE_SIZE / 2 - 50, 100, 10, buttons & 1 ? BLUE : DARKBLUE);
53     DrawCircle(center_x - SQUARE_SIZE / 2 + 50, SQUARE_SIZE, 10, buttons & 4 ? GOLD : YELLOW);
54     DrawCircle(center_x + SQUARE_SIZE / 2 - 50, SQUARE_SIZE, 10, buttons & 2 ? GOLD : YELLOW);
55
56     DrawCircle(200 - 60, SQUARE_SIZE / 2 + 60, 60, IsKeyDown(KEY_TESLA_MODE) ? YELLOW : BLUE);
57 }

```

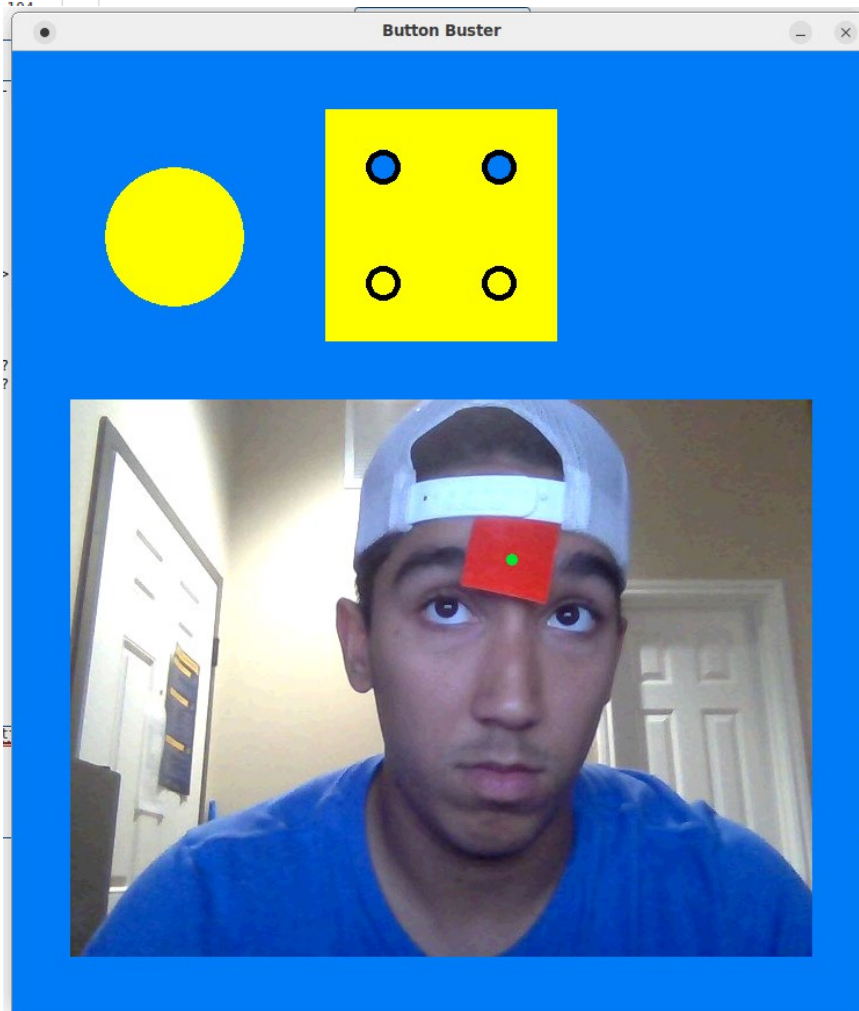


2. Modify the code to track an object only when the T key is pressed

```

79     buttons = (IsKeyDown(KEY_TESLA_MODE) ?
80               ((centroid_x > 640 / 2 + buffer) ? ((centroid_y < yMid && centroid_y > frame_y) ? 4 : 8) : 0 +
81               (centroid_x < 640 / 2 - buffer && centroid_x > frame_x) ?
82               ((centroid_y < yMid && centroid_y > frame_y) ? 2 : 1) : 0 +
83
84               ((centroid_y < yMid && centroid_y > frame_y) ?
85               ((centroid_x < 640 / 2 + buffer) && (centroid_x > 640 / 2 - buffer) ? 6 : 0) :
86               ((centroid_x < 640 / 2 + buffer) && (centroid_x > 640 / 2 - buffer) ? 9 : 0)) : 0) +
87
88               (IsKeyDown(KEY_LEFT_FORWARD) * 8) +
89               (IsKeyDown(KEY_LEFT_BACK) * 4) +
90               (IsKeyDown(KEY_RIGHT_BACK) * 2) +
91               (IsKeyDown(KEY_RIGHT_FORWARD) * 1) +
92
93               gpio_get_buttons();
94
95     set_pwm5(buttons, current_speed);
96
97     draw(buttons);
98
99     EndDrawing();
100
101     do_unload_frame();
102
103
104

```



6. Day Dream

- Make tesla mode toggleable, so you don't need to keep holding the key
- Make it so that it can detect how far the red tape is so that it can just move towards it
- Ability to detect and avoid obstacles while following the red tape, like auto correction in a tesla

7. Problems experienced and areas of concern

- Memory was not being cleared causing pi to crash about every 30 seconds, making it difficult to work
 - Was not clearing memory with `do_unload_frame()`
- Lights were not coming on pi display, but were turning on protoboard
 - Had old code to test the program that would set buttons to 0 causing flickering

8. Course goals

Rank the level (0=none to 5=a large amount) that this lab covered the each of the course goals by updating the appropriate box in the table below.

Course Goals	Laboratory												
	01	02	03	04	05	06	07	08	09	10	11	12	13
To compile, link, debug and run C programs	4	5	5	5	5	5	5						
To use the basic C program structure; variables, constants, and operators	4	4	4	4	4	5	5						
To use the repetition constructs such as looping with for, while, and do-while statements	2	3	4	4	4	4	4						
To use the selection structures such as if, if/else, switch, conditional expression statement	0	0	3	3	3	3	4						
To create program modules using functions (passing data to and returning values from functions)	0	0	0	2	4	4	4						
To use arrays and pointers	0	0	0	0	4	3	4						
To work with strings and string manipulating functions	2	4	4	1	2	1	1						
To perform file, I/O operations	3	4	4	5	5	5	5						
To develop structured, modular, and top-down design of software	0	0	0	1	1	4	5						